

Smart MCQ Practice App – Project Report

Project Overview

Smart MCQ Practice App is an Android-based adaptive learning application developed using Kotlin and Android Studio. The application helps users practice multiple-choice questions while improving their learning efficiency through the implementation of a Spaced Repetition Algorithm.

The application provides an interactive practice environment, tracks user performance, and presents analytical insights through a dashboard.

Implemented Features

The following features have been implemented:

1. Splash Screen

- Displays application branding.
- Provides smooth transition to the home screen.

2. Home Screen

- Welcome card.
- Motivational hero section.
- Start Practice navigation card.
- Dashboard navigation card.
- Performance trend chart.

3. Practice Module

- Displays MCQ questions.
- Multiple-choice answer selection.
- Immediate feedback for correct and incorrect answers.
- Progress tracking.
- Toolbar navigation.

4. Dashboard Module

- Total questions attempted.
- Total correct answers.
- Total incorrect answers.
- Accuracy percentage.
- Learning improvement tracking.

- Progress indicators.

5. Local Database

- Room Database integration.
- Persistent storage of quiz attempts.
- Retrieval of historical performance data.

6. User Interface

- Material Design 3 implementation.
- Custom color themes.
- Card-based layouts.
- Smooth screen transition animations.

How Spaced Repetition Is Applied

The application uses a simplified Spaced Repetition algorithm to improve learning retention.

The algorithm prioritizes questions according to the user's previous performance:

- Correctly answered questions receive lower priority.
- Incorrectly answered questions receive higher priority.
- Questions answered incorrectly are reinserted into the practice queue.
- Users therefore practice difficult questions more frequently.

This adaptive learning approach helps users strengthen weak areas while reducing repetition of already mastered questions.

Dashboard Overview

The dashboard provides analytical insights into user performance.

It includes:

- Total attempted questions.
- Number of correct answers.
- Number of incorrect answers.
- Overall accuracy percentage.
- Learning improvement indicator.
- Performance visualization through charts.

The dashboard allows users to monitor their progress and evaluate their learning effectiveness over time.

Technologies Used

- Kotlin
- Android Studio
- Room Database
- Kotlin Coroutines
- Material Design 3
- MPAndroidChart
- Android SDK

Conclusion

Smart MCQ Practice App demonstrates the implementation of adaptive learning principles, local data persistence, performance analytics, and modern Android application development practices. The project successfully combines educational technology concepts with a user-friendly mobile interface.